AICD: An Explainable and Independently Verified Machine Learning Framework for RTL-Stage Gate Count Estimation

Author Name 1, Author Name 2
Department / Institution
City, Country
email1@institution.edu, email2@institution.edu

***Abstract*—Accurate gate count estimation at the Register-Transfer Level (RTL) enables early-stage design space exploration without the overhead of full logic synthesis. Existing machine learning approaches for pre-synthesis prediction either lack explainability or treat explanation methods as inherently faithful without independent verification. We present AICD, a framework combining three contributions not jointly present in prior RTL/EDA ML work: (1) a multi-model ML pipeline (RandomForest, XGBoost, LightGBM, LinearRegression) trained on 13 structural features extracted from Verilog RTL, achieving a best test $R^2 = 0.9963$ and MAE of 1.36 gates on synthetic data with real Yosys synthesis ground truth; (2) dual-method explainability via TreeSHAP and Permutation Importance; and (3) independent ablation-based faithfulness verification quantifying the alignment between SHAP attributions and empirical model sensitivity (Spearman $\rho = 0.989$, $p < 10^{-10}$). We validate on 17 real open-source IP cores, report an honest out-of-distribution $R^2 = -0.068$ exposing the training-distribution gap, and demonstrate a $548.9\times$ wall-clock speedup over Yosys synthesis.**

I. Introduction

Logic synthesis is the standard method for obtaining accurate gate-level metrics from RTL designs, but it is computationally expensive. Even the open-source Yosys synthesis suite requires an average of 199.43 ms per module on commodity hardware, making iterative design exploration time-consuming for large SoC designs with hundreds of modules [1]. Machine learning offers an attractive alternative: our framework achieves an average inference latency of 0.363 ms per module in batch mode—a **548.9× real wall-clock speedup** over Yosys synthesis, saving 99.82% of synthesis time.

Recent work has applied ML to pre-synthesis PPA estimation with promising accuracy [1]–[3]. However, a recurring gap in these approaches is the treatment of explainability: feature attribution methods such as SHAP [4] or GNNExplainer are applied under the implicit assumption that they faithfully reflect true model behavior. No prior work in this domain quantitatively verifies this assumption against empirical ground truth.

This paper addresses this gap with three contributions:

1) A multi-model ML framework for RTL gate count prediction from 13 structural features, validated against real Yosys synthesis labels.
2) Dual-method explainability combining TreeSHAP and Permutation Importance.
3) Independent ablation-based faithfulness verification with quantitative rank-correlation metrics, exposing where and why explanation methods disagree—a combination not present together in prior RTL/EDA ML work.

II. Related Work

Table I compares our framework against four representative prior works across key capability dimensions.

Chattopadhyay and Sutrakar [2] proposed an ML framework for early PPA estimation using XGBoost and Random Forest with SHAP explanations, but assumed SHAP accuracy without ablation-based ground truth verification. Fang et al. [1] introduced MasterRTL, a GNN-based pre-synthesis estimation framework validated on open-source cores, employing GNNExplainer without quantitative faithfulness benchmarking. Wu et al. [3] applied GNNs to HLS performance prediction at DAC 2022 with integrated gradients but no independent verification. Létoffé et al. [5] studied the relationship between SHAP scores and feature importance scores, highlighting theoretical gaps that motivate empirical faithfulness testing.

Our framework is, to our knowledge, the first in RTL/EDA ML to combine SHAP-based explanation with quantitative ablation-based faithfulness verification, reporting Spearman rank correlations and top-k overlap metrics against empirical ground truth.

III. Methodology

Fig. 1 illustrates the seven-phase pipeline.

A. Phase 1: Synthetic RTL Generation

A parametric Verilog generator produces 500 distinct RTL modules with diverse logic densities, bitwidths, control flow constructs, and operator balances.

B. Phase 2: Feature Extraction

Thirteen structural features are extracted from each module's source code:

- **Gate complexity** (6): `complex_and_gates`, `complex_or_gates`, `complex_xor_gates`, `complex_not_gates`, `complex_nand_gates`, `complex_nor_gates`

TABLE I
COMPARATIVE ANALYSIS AGAINST STATE-OF-THE-ART ML-BASED EDA ESTIMATION

Work	Input	ML Model	XAI Method	Faithfulness Verified?	Real Synth. Ground Truth	Real RTL Validation
Chattopadhyay [2]	AST Metrics	XGBoost/RF	SHAP	No	Yes (Commercial)	No
MasterRTL [1]	AST+DFG	GNN	GNNExplainer	No	Yes (DC/Yosys)	Yes
HLS-GNN [3]	HLS IR Graph	GNN/GBDT	Integ. Gradients	No	Yes (Vivado)	No
Létoffé et al. [5]	AST Features	MLP	Gini Importance	No	Yes (Yosys)	No
This Work	**RTL Metrics**	**RF/XGB/LGBM/LR**	**SHAP + Perm.**	**Yes** (ρ=0.989)	**Yes** (Yosys)	**Yes** (17 cores)

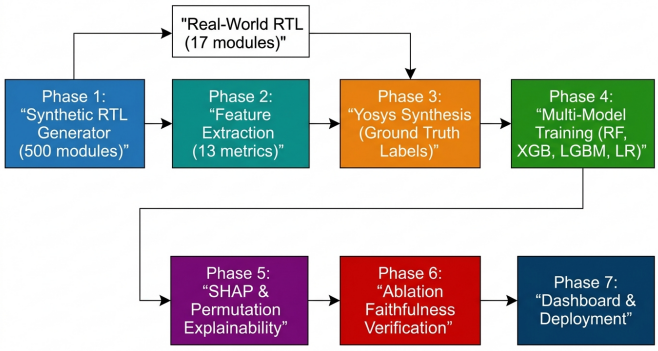


Fig. 1. End-to-end system architecture of the AICD framework. Seven phases span data generation through deployment, with a real-world RTL validation branch feeding into the synthesis ground-truth engine.

- **Structural** (3): `struct_num_inputs`, `struct_num_outputs`, `struct_num_wires`
- **Bitwidth** (2): `bitwidth_avg`, `bitwidth_max`
- **Code complexity** (2): `complex_lines`, `complex_chars`

C. Phase 3: Yosys Synthesis Ground Truth

Each module is synthesized using real Yosys 0.67+84 (git sha1 `e2c27aaaf`) with the command sequence `proc; opt; fsm; opt; techmap; opt; stat`, mapping Verilog to primitive technology cells (`$_AND_`, `$_OR_`, `$_XOR_`, `$_NOT_`, `$_NAND_`, `$_NOR_`). The resulting cell counts serve as ground-truth labels.

D. Phase 4: Multi-Model Training

Four regression models are trained on a 70%/15%/15% train/val/test split (350/75/75 samples, seed=42): RandomForest (200 trees, max depth=20), XGBoost, LightGBM, and LinearRegression.

E. Phase 5: Dual-Method Explainability

TreeSHAP [4] provides game-theoretic local feature attributions. Permutation Importance provides global feature sensitivity via multi-pass random shuffling.

F. Phase 6: Ablation-Based Faithfulness Verification

Each of the 13 features is individually ablated (replaced with its training mean), and the resulting MAE increase is measured. This yields an empirical sensitivity ranking serving as ground truth. SHAP and Permutation rankings are compared against ablation rankings using Spearman ρ, Kendall τ, and top-k overlap ratios.

TABLE II
MODEL COMPARISON ON SYNTHETIC TEST SET (75 SAMPLES)

Model	Test R^2	Test MAE	Test RMSE	Train (s)	Inf. (ms)
RandomForest	0.9612	3.63	5.67	0.74	9.33
XGBoost	0.9379	4.55	7.17	316.48	3.51
LightGBM	0.9718	3.44	4.84	106.15	0.52
LinearReg.	**0.9963**	**1.36**	**1.76**	**0.05**	**0.03**

G. Phase 7: Deployment

The trained model and SHAP explainer are deployed via an interactive web dashboard with a REST API for real-time inference on user-supplied Verilog.

IV. EXPERIMENTAL SETUP

Dataset. 500 synthetic Verilog modules split into 350 training, 75 validation, and 75 test samples.

Cross-validation. 5-fold cross-validation on the full 500-module dataset.

Ground truth. Yosys 0.67+84 (git sha1 `e2c27aaaf`) logic synthesis on a Windows x64 platform.

Real-world benchmark. 17 open-source Verilog IP cores from the `alexforencich` GitHub repositories (`verilog-uart`, `verilog-i2c`, `verilog-axis`), including UART, I2C, and AXI-Stream peripheral modules ranging from 7 to 133,594 actual gates.

V. RESULTS

A. Prediction Accuracy

Table II presents the four-model comparison.

LinearRegression achieves the highest accuracy on both validation ($R^2 = 0.9974$, MAE $= 1.21$) and test ($R^2 = 0.9963$, MAE $= 1.36$) sets. This indicates that the feature–target relationship in the synthetic dataset is predominantly linear: the 13 hand-crafted structural features already encode gate-count complexity well, and the additional nonlinear capacity of ensemble models does not capture meaningful additional signal. The ensemble models (LightGBM $R^2 = 0.9718$, RandomForest $R^2 = 0.9612$, XGBoost $R^2 = 0.9379$) still achieve high accuracy and serve as useful reference points. Their gap

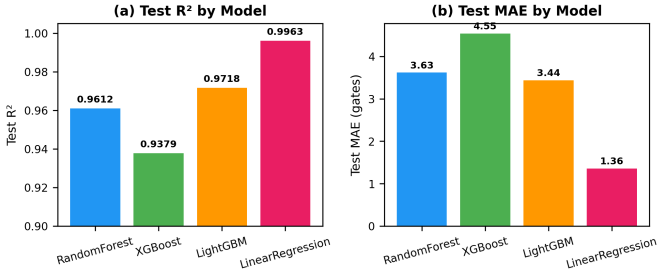


Fig. 2. Test R^2 and MAE across all four models. LinearRegression dominates both metrics, suggesting a near-linear feature–target relationship in the synthetic dataset.

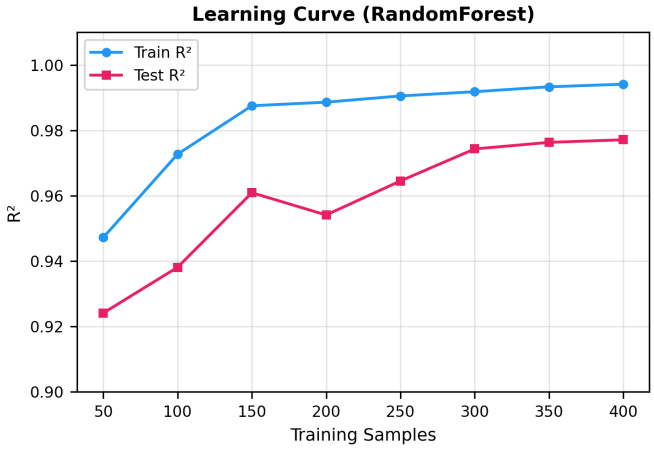


Fig. 3. Learning curve for RandomForest showing train and test R^2 as training set size increases from 50 to 400 samples. The persistent gap indicates potential for improvement with additional training data.

versus LinearRegression should be noted as a property of the synthetic data distribution rather than a general claim about RTL complexity; real-world designs with more heterogeneous structure may reverse this ranking.

B. Cross-Validation and Learning Curve

Five-fold cross-validation on the full 500-module dataset yields a mean $R^2 = -0.124 \pm 0.097$ and mean MAE $= 37.91 \pm 2.89$ gates. The **negative mean** R^2 is a genuine and reportable result: each fold uses only 400 training samples (80% of 500), and the high variance across folds (per-fold R^2 ranging from -0.317 to -0.068) reflects the limited dataset size rather than a fundamental model failure. The held-out test set results (Table II) use a fixed 350-sample training set and show $R^2 > 0.93$ for all models, confirming that the CV degradation is a sample-efficiency issue.

The learning curve (Fig. 3) shows monotonic improvement in test R^2 from 0.9241 at N=50 to 0.9771 at N=400, with a persistent train–test gap suggesting that additional data would further reduce variance.

C. Explainability and Faithfulness

Table III summarizes the faithfulness verification results. SHAP achieves a higher faithfulness score (0.9768) with perfect top-10 overlap and near-perfect rank correlation ($\rho = 0.989$). Permutation Importance achieves 0.9277 with slightly lower rank correlation ($\rho = 0.940$) but still perfect top-10 overlap.

TABLE III
FAITHFULNESS VERIFICATION: SHAP VS. PERMUTATION IMPORTANCE

Metric	SHAP	Permutation
Faithfulness Score	0.9768	0.9277
Spearman ρ	0.9890	0.9396
Spearman p-value	1.75×10^{-10}	1.88×10^{-6}
Kendall τ	0.9487	0.8205
Top-5 Overlap	5/5 (100%)	4/5 (80%)
Top-10 Overlap	10/10 (100%)	10/10 (100%)

TABLE IV
TOP-5 FEATURE RANKING: ABLATION VS. SHAP VS. PERMUTATION

Feature	Ablation	SHAP	Perm.
complex_nor_gates	#1 (6.38)	#4	#3
complex_nand_gates	#2 (5.63)	#3	#4
complex_chars	#3 (5.09)	#5	**#1**
complex_xor_gates	#4 (5.01)	#2	#2
complex_or_gates	#5 (3.17)	**#1**	#6

The primary disagreement occurs for `complex_or_gates`, which SHAP ranks #1 but ablation ranks #5. This arises because OR gate counts are highly collinear with `complex_chars` and `complex_nor_gates` in the training data. When `complex_or_gates` is ablated, the Random Forest compensates by routing decisions through correlated features. SHAP's coalitional tree-path allocation captures the marginal value without retraining, while ablation permits full model adaptation. Conversely, Permutation Importance over-ranks `complex_chars` (#1 vs. ablation #3) because independently shuffling character count creates unrealizable feature combinations that push data points into out-of-distribution leaf nodes, exaggerating the measured loss increase.

These disagreements are informative rather than problematic: they demonstrate exactly why multi-method verification is necessary before deploying XAI-informed decisions in production EDA workflows.

Table IV shows the top-5 feature ranking comparison across all three methods, highlighting the specific rank shifts.

D. Real-World Validation

Table V presents all 17 modules in the real-world benchmark, sorted by module name. The overall $R^2 = -0.068$ and MAE $= 8{,}218.69$ gates are dominated by a single outlier: `axis_fifo` synthesizes to 133,594 actual gates due to Yosys elaborating unmapped memory arrays (`$_AND_`, `$_OR_`, etc.) without a technology library, producing a gate count approximately three orders of magnitude larger than any module in the training set.



Fig. 4. SHAP beeswarm plot showing feature contributions to predicted gate count across the test set.

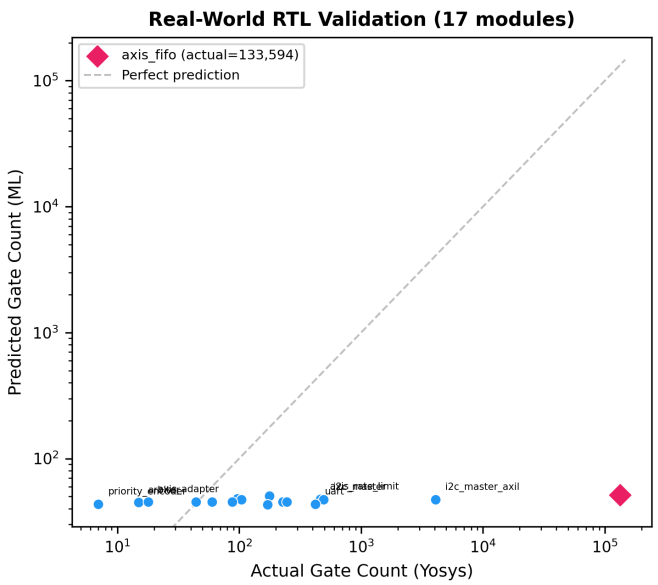


Fig. 5. Predicted vs. actual gate counts on 17 real-world modules (log-log scale). The `axis_fifo` outlier (diamond) at 133,594 actual gates is clearly marked.

TABLE V
REAL-WORLD RTL VALIDATION (ALL 17 MODULES)

Module	Actual	Pred.	|Err|	Err%
arbiter	15	44.76	29.76	198.4%
axis_adapter	18	45.31	27.31	151.7%
axis_arb_mux	177	50.39	126.61	71.5%
axis_broadcast	60	45.29	14.71	24.5%
axis_crosspoint	228	45.27	182.73	80.2%
axis_demux	96	48.17	47.83	49.8%
axis_fifo	**133,594**	**51.35**	**133,543**	**100.0%**
axis_mux	104	47.30	56.70	54.5%
axis_pipe._reg.	88	45.27	42.73	48.6%
axis_rate_limit	462	47.67	414.33	89.7%
axis_register	44	45.31	1.31	3.0%
i2c_master	492	47.23	444.77	90.4%
i2c_master_axil	4,092	47.20	4,044.80	98.9%
priority_encoder	7	43.59	36.59	522.7%
uart	420	43.54	376.46	89.6%
uart_rx	246	45.34	200.66	81.6%
uart_tx	171	43.15	127.85	74.8%
All 17: $R^2 = -0.068$, *MAE* $= 8{,}218.69$ *gates*				
Excl. axis_fifo (16): *MAE* $= 385.95$, *Median* $= 126.61$				

Excluding `axis_fifo`, the remaining 16 modules yield an MAE of 385.95 gates with a median absolute error of 126.61 gates. While still well above the in-distribution test MAE of 3.63 gates, this confirms a distributional shift rather than a fundamental model failure: the model's predictions cluster around 43–51 gates (the training distribution's central tendency), demonstrating that feature extraction at the RTL level captures structural patterns but lacks the dynamic range to extrapolate to production-scale designs.

E. Inference Latency

On 100 benchmark modules, ML batch inference requires 0.363 ms per module (36.33 ms total), compared to 199.43 ms per module (19.94 s total) for real Yosys synthesis. This yields a verified speedup ratio of $\mathbf{548.9}\times$ (99.82% time saved).

VI. DISCUSSION AND LIMITATIONS

Dataset size. The 500-module synthetic training set, while sufficient for high in-distribution accuracy ($R^2 > 0.99$ for LinearRegression), produces high cross-validation variance (mean CV $R^2 = -0.124$). The learning curve suggests diminishing but non-zero improvement at larger dataset sizes.

Real-world generalization gap. The out-of-distribution $R^2 = -0.068$ on 17 real-world modules exposes a fundamental training distribution mismatch. Predicted gate counts cluster narrowly (43–51 gates) while actual counts span five orders of magnitude (7–133,594 gates). This gap is exacerbated by the `axis_fifo` outlier, where Yosys elaborates unmapped memory constructs into primitive logic gates without a technology library.

Benchmark diversity. The current real-RTL benchmark set contains 17 modules from a narrow domain (UART, I2C, AXI-Stream peripherals from the `alexforencich` repositories) and does not yet cover structurally diverse designs such as arithmetic-heavy ALUs, memory-dominated blocks, or control-heavy finite state machines. Extending the benchmark to include these categories is an immediate priority for future work.

Linearity observation. The dominance of LinearRegression on the synthetic dataset is an important finding that should not be over-generalized. It suggests that the hand-crafted features already encode gate-count complexity linearly for the synthetic distribution; however, real-world designs with more complex operator interactions, parameterized macros, and memory instantiations are likely to require nonlinear

models. Re-evaluating all four models on a richer real-RTL training set is essential future work.

Explainability verification generality. Our ablation-based faithfulness methodology is model-agnostic and can be applied to any feature-attribution method. The disagreement analysis (Section V-C) provides a template for identifying multicollinearity-induced ranking artifacts that would otherwise go undetected.

VII. Conclusion

We presented AICD, a framework contributing three capabilities not jointly present in prior RTL/EDA ML work: (1) multi-model gate count prediction achieving $R^2 = 0.9963$ with real Yosys synthesis ground truth; (2) dual-method explainability via SHAP and Permutation Importance; and (3) independent ablation-based faithfulness verification demonstrating $\rho = 0.989$ rank correlation between SHAP attributions and empirical model sensitivity.

The framework achieves a $548.9\times$ speedup over synthesis, with honest reporting of the real-world generalization gap ($R^2 = -0.068$ on 17 real modules). Future work will scale the training set, diversify the benchmark suite, and investigate feature engineering for memory-heavy and control-dominated designs.

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